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Lesson Notes

MODULE 4 | LESSON 4

OPTION STRATEGIES AND SCENARIOS

Reading Time	20-25 minutes
Prior Knowledge	Option Payoffs, Leverage, Stocks, Calls, Puts, Non-linearity, In the money (ITM), At the money (ATM), Out the money (OTM)
Keywords	European Style of Exercise, American Style of Exercise

In the previous lesson, we showed that non-recourse mortgaged homes resembled equity options in their non-linearity and overall payoff. In this lesson, we discuss some of the strategies and scenarios with options.

1. Options: Exercise Type

There are three types of exercise methods for options:

1. European options can only be exercised at the option's expiration.
2. American options can be exercised throughout the option's lifetime, including expiration.
3. Bermudan options are like American options in that they can be exercised early but only on selected dates. We will discuss Bermudan options in a much later course.

For now, let's simply consider European-style options versus American-style options. Which of these sounds more flexible? The American exercise. Indeed, the American-style option is a

superset of the European option, in that it contains the extra optionality to exercise early.

For puts especially, this early exercise feature can be valuable. Suppose you are long a put option at a strike of \$50. Suppose that the stock price went to nearly zero. If you held a European put, you could not do anything until the expiration of the option. Perhaps the stock price could rebound and then your put could actually expire worthless. For the American put, however, you would be able to exercise early, collecting $K - 0$, or K dollars. You can then take those K dollars that you would earn per share (times 100 shares) and invest in a risk-free account for the remainder of the option's life, further increasing the amount that you would earn. Early exercise can be optimal for puts, depending on its price path. In particular, for a put that is very deep in the money, the put's payoff can be near its maximum, which is simply K . You will see in a future course the details about the pricing and hedging of puts and calls, as well as the times in which it makes sense to exercise early. But for now, let's add the exercise style as another item to our list of decisions to make when selecting an option. We'll continue that in the next section.

2. Option Strikes

In the previous section, we discussed the different styles of exercise. We can include a few additional elements to the list of items that one should consider when trading options.

1. Strike price
2. Direction
3. Timing
4. Size

For the direction, we have two choices: calls or puts (assuming we are not volatility traders).

For timing, we have several choices: 1-month, 2-month, 3-month, ..., perhaps up to 12 months.

For strike, we have numerous choices: dozens of possibilities at \$2.5 increments above and below the current stock price.

Choosing the option's strike is likely the most difficult choice to make. Choosing the strike is effectively choosing the size move

within the option's lifetime. The higher the strike, the less expensive the call, but choosing too high a strike may result in the option never getting in the money. To recap, let's say you bought a one-month call at a strike of 110, when the price is 100. In one month's time, the final stock price is 109. Options struck at 102.50, 105, 107.50 are all in the money, but the option struck at 110 is not. There is no partial credit for getting the direction (call) and timing (one month) parts right if the size (strike level) is wrong.

There are three ways we can categorize call options by their strikes:

1. **In-the-money** calls: Strikes are less than the current stock price
2. **At-the-money** calls: The strike closest to or equal the current stock price
3. **Out-of-the-money** calls: Strikes that are greater than the current stock price

For puts, we have the same categories but need to apply the put's payoff.

1. **In-the-money** puts: Strikes are greater than the current stock price
2. **At-the-money** puts: The strike closest to or equal the current stock price
3. **Out-of-the-money** puts: Strikes that are less than the current stock price

In-the-money options are known as ITM. At-the-money options are abbreviated as ATM. Out-of-the-money options are abbreviated as OTM.

Now, ITM options (both calls and puts) are more expensive than ATM options, because they have greater intrinsic value. Both these options are more expensive than OTM options for the same reason. There is no free lunch. If you buy an OTM option, it is less expensive but has high risk. Only options that expire ITM will have positive payoffs.

What specifically happens at strike? As we know from the option's payoff, the payoff slope abruptly changes from flat to upward-sloping. If you were holding an ATM option, you would like the stock's price to increase so it becomes ITM. However,

the party who sold the option would like just the opposite, making the stock's price decrease so it is OTM. Indeed, there are often "tug-of-war" games that occur at strike levels to fight for options going in and out of the money.

Well, why buy an option in the first place? Let's categorize three types of traders: hedgers, speculators, and arbitrageurs.

3. Hedging with Options

Let's consider the trader who wishes to minimize risk with options. We'll consider risk as volatility. We would like to minimize the standard deviation (or variance) of outcomes. Suppose you're a farmer and you need to purchase seed for this year's harvest. You're willing to pay \$50 a bushel for seed. If you were to pay more, you're unsure that you could sell it at a price that would keep the farm profitable. Therefore, you want to lock in the price of \$50. You might first consider going to the futures market. The futures contract would cost nothing to enter. Perhaps you could find the price at \$50 bushel in three months' time. But what if the price of the seed dropped over the next three months to \$30? Your competition could buy it at that lower price, and sell their crops at lower rates than you because you were locked in at \$50. Your higher costs would inevitably result in higher prices, weakening your competitiveness on price. Instead of a future, what if you had bought a call option? Compared to futures, options look expensive. The futures require no upfront capital but could lead to very unfavorable conditions. Although options have a premium, they never lead to trading at unfavorable prices. Suppose instead you were to purchase a call option struck at 50 to buy seed. One of two scenarios could occur.

In scenario 1, the price of seed in three months goes well above 50. Having the option struck at 50, you have locked in that price and will buy the seed at a level much lower than the market. Later, you will be able to sell the crops at a very competitive price since your costs were relatively low.

In scenario 2, the price of seed in three months goes below 50. Having an option to buy at 50 is unhelpful, so you simply don't exercise it. You buy your seed at the lower price along with everyone else. The option served as an insurance against seed inflation.

When options are used to minimize risk, you can think of them as insurance. When futures are used, they can help to minimize risk, but they can create unfavorable terms due to price changes between the locked-in future price and the actual future price. Options minimize risk more because the premium is the maximum amount you could lose. If the option is exercised, its payoff is positive, and you can get a better price using the strike level than the current market. If the option is not exercised, it's because prices decreased, and the option provided a hedge against a price increase. The difference between futures and options is that options provide a fixed cost (the premium) paid in the present, while futures have a variable cost (the difference between the locked and market prices) paid in the future.

We've given an example of someone who's buying a call to minimize risk by guaranteeing a maximum price for their inputs. Similarly, we could construct an example of someone buying a put to minimize risk by guaranteeing a minimum price for their outputs. The same farmer might purchase a put to sell corn at \$90 a bushel. For the cost of a premium, the farmer guarantees a sale at 90 even if it drops in the future, while still having the flexibility to get an even higher price than 90 if future prices increase.

4. Speculating with Options

We've given the example from a risk-management point of view. But indeed options can work the other way if you are speculating. A speculator is betting on a market direction and uses derivatives for their leverage, for example, if you were very bullish on a stock. Consider two different options strategies you could consider.

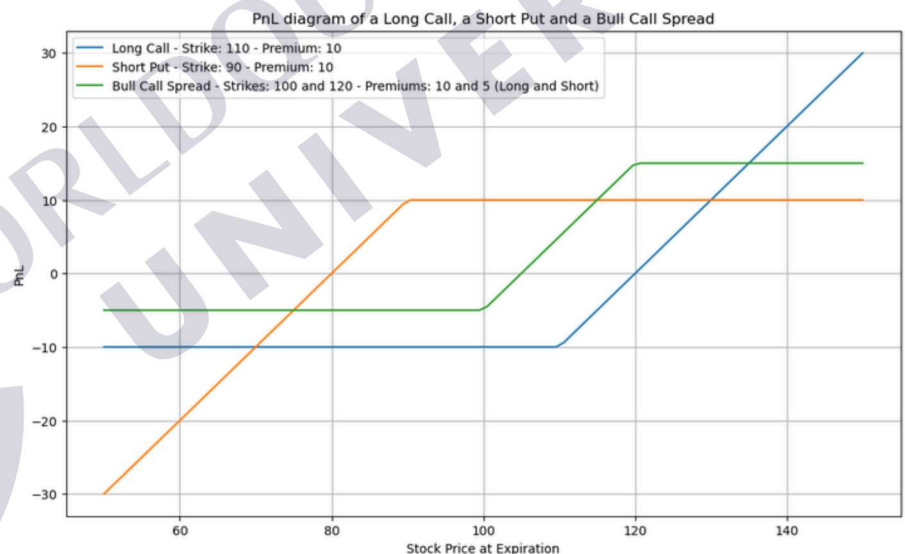
In strategy 1, you would simply buy calls. You have to pay the cost of the premium (with no financing). If the stock price goes up, within the option's time frame, and past the strike you chose, then the option is ITM. Of course, you have to pick the right strike and the right expiration date.

In strategy 2, you simply sell puts. Selling anything generates cash flow. Selling a put means you are exposed if the stock price drops. But if you strongly believe the stock will go up, you make money, within the option's time frame, so long as it ends up higher than the put's strike. This strategy does not require

any upfront capital (except that if the price subsequently drops, then some cash will be needed to satisfy margin requirements).

In strategy 3, you buy a call at a strike, say 100, and then sell a call at a higher strike, say 110. This is known as a bull spread strategy. This means that you're paying money for the first call but you're actually collecting money for the second call because it's getting sold. Compared to strategy 1, this strategy is cheaper because you are "refunded" the cost of the second option. However, this limits the upside of your investment to the difference between the two strikes: in our example, \$10. If you wanted to have a higher potential profit, then you could sell the second option at a higher strike. Strategy 3 is known as a bull spread. At the second strike, the two calls offset each other, so the P&L comes from the first call being $(K_2 - K_1)$ dollars in the money. The bull spread is appropriate if you have "bounded" optimism rather than "unbounded" optimism about the stock's upside.

Here are the three speculative strategies compared:



The put and bull spread have less upside than the call. The put's upside is maxed at its strike, K . The bull spread's upside is maxed at the difference between its two strikes.

Let's compare some of these strategies. In this working example, you select options: calls or puts, long or short, and their strikes. They all have the same underlying and expiration. This graph shows the option payoffs, and where the options will be in-the-money and out-of-the-money.

For either hedgers or speculators, options can expire worthless. The difference for a hedger though is that the option not exercised leads to more favorable conditions (e.g., buying at a lower price). For the speculator, the worthless option is a capital loss with no favorable conditions. Since options can lose 100% of their value, they are intended to be used with great care. A buy-and-hold strategy that could work for many stocks or bonds is likely going to be inappropriate for options.

Recall that in Module 1 we looked at investments where your interest and principal could be protected. Equities eliminated the protection but still offered some dividends and ownership of assets should a company liquidate. Cryptos offered less than equities, but did not have an expiration date. ETFs provided protection through diversification at one level or another. Options are in their own class. As derivatives, they inherently have both leverage and non-linearity. The combination comes with potentially catastrophic results if one does not understand the risks.

For hedgers, the risk is similar to the cost of insurance, but for speculators, it is the entire principal. Let's say the hedger's option expires worthless. It can be thought of as the cost of insurance (like fire insurance that provides peace of mind). The fact that the hedger's option is OTM means that there's a favorable condition, such as being able to buy at an even lower price than the strike offered. This situation is most likely preferable for the hedger. Now let's say the speculator's option expires worthless. There's no other favorable condition. There is no analogy to insurance because risks were being increased, not decreased. Speculate carefully! Options can and do lose 100%. The difference between a payoff and no payoff is simply one point: the strike level.

5. Seeking Arbitrage with Options

We'll end with a brief but exciting type of option trader: the arbitrageur. The arbitrageur could buy calls, puts, stocks, and bonds across different exchanges at the same time if the prices of equivalent payoffs differ, thereby earning a riskless profit. How? Equity options are combinations of stocks and bonds. It is possible to replicate an option with some combination of stocks and bonds. There is a relationship among calls, puts, stocks, and bonds that shows a particular

combination of some equals a particular combination of the other. If those costs are different, an arbitrageur could sell the "expensive" one, buy the "cheap" one, pocket the difference, and the cash flows net out without any further expenses. The arbitrageur locks in a profit through a fleeting and infrequent situation that would create the opportunity.

Consider two different portfolios:

1. Portfolio A is long a call and short a put. These options are struck at K and expire at time T .
2. Portfolio B is long a stock and short a bond; the bond has notional value K , earns the risk-free rate r , and matures at time T .

Note that the bond's price at the present time is $K * e^{-rt}$

There are three states where the stock price can finish: smaller than the strike, equal to the strike, and greater than the strike.

If the options wind up ATM, then the call and put have no value. Otherwise, only one of them can end up ITM.

If $S < K$, then the put is ITM; since the position is short the put, the payoff is $-(K - S)$ which is $S - K$.

If $S > K$, then the call is ITM, which is $S - K$.

If $S = K$, then we are ATM, at which point $S - K = 0$, but we can leave it as $S - K$.

So all three outcomes show that the final payoff is $S - K$.

This matches the payoff of buying the stock and selling the bond.

	$S < K$	$S = K$	$S > K$
call - put	$(S - K)$	$0 = (S - K)$	$S - K$
stock - bond	$S - K$	$S - K$	$S - K$

Suppose the cost of Portfolio A were \$8.

Suppose the cost of Portfolio B were \$5.

Do you see an arbitrage?

Sell Portfolio A, buy Portfolio B, and collect \$3 ($8 - 5$). That is pure profit, as the two portfolios offset each other.

To be sure, this type of arbitrage would be conducted in fractions of a second to discover and execute across three different markets: bonds, stocks, and options. Although you could "try this at home," the opportunities will be gone if you blink. Nevertheless, this idea of arbitrage, or rather the absence of it, will be vital when we relate the derivative's price back to the underlying. That will be the topic of a later course. For the next module, we'll revisit mortgages and see the role they play in the housing market.

6. Conclusion

In this lesson, we wrapped up our discussion of options by detailing some of the strategies and scenarios. In the next module, we extend our coverage of financial instruments to the world of securitization.

